solubility of neptunium in all oxidation states seems to be limited by the low solubility of $Np(OH)_4$ (am) or NpO_2(c).

In oxic waters, uranium is present as U(VI) and strongly complexes with carbonate; e.g. in sea water, the uranium is present at 10^{-8} M concentration as $UO_2(CO_3)_3^{4-}$. The solubility of uranium in some waters may be limited by the formation of an uranyl silicate species.

22.7. SPECIATION CALCULATIONS

An essential step in the safety analysis of potential waste repositories is the prediction of what chemical species are formed in the actual water. For example, the relatively high solubility of uranium in sea water is due to this strong carbonate complexation which forms $UO_2(CO_3)_3^{4-}$. Figure 22.8 shows the variation of uranyl species in a surface water under normal atmospheric pressure of CO_2 ($p_{CO_2} \sim 3.2 \times 10^{-4}$; $\log[CO_3^{2-}] = 2pH - 18.1 + \log(p_{CO_2})$). These speciation diagrams are calculated from the equilibrium constant for formation of each species plus mass balance equations. In this section we describe the use of equilibrium constants in modeling the speciation in a natural water.

Assume the reactions $M + nX = MX_n$ where X is OH^-, and CO_3^{2-} and $n = 1$ to 3. The equilibrium constants (assuming for simplicity that the chemical activity coefficients all are unity), expressed as β are given by (cf. eqn. (18.31)):

$$\beta_n = [MX_n]/[M]\,[X]^n \tag{22.3}$$

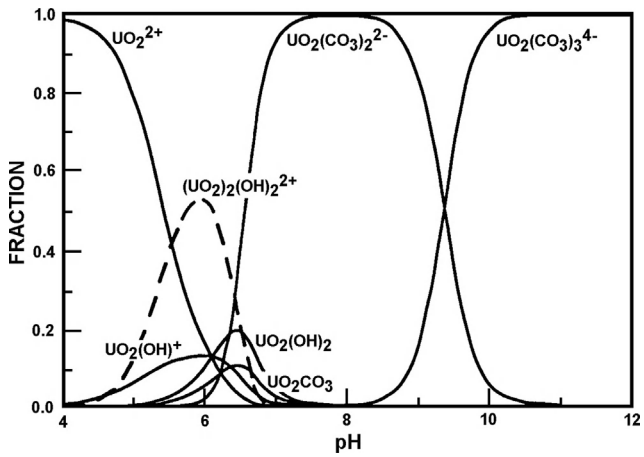


Figure 22.8 Fraction of uranium species in water with natural carbon dioxide content and at different pH, showing hydrolysis and CO_3^{2-} complexation.

The equilibrium constants for Am(III) are listed in Table 22.9 for hydrolysis and CO_3^{2-} complexation. Let us consider the case of Am(III) should it be released into the environment.

The first step in modeling the speciation of Am(III) is to rearrange the above equation to express the ratio of complexed to free metal, e.g. (omitting ionic charges)

$$[\mathrm{AmX}]/[\mathrm{Am}] = \beta_1[\mathrm{X}] \tag{22.4}$$

$$[\mathrm{AmX_2}]/[\mathrm{Am}] = \beta_2[\mathrm{X}]^2 \tag{22.5}$$

and so forth. The mass balance equation is:

$$[\mathrm{Am}]_\mathrm{T} = [\mathrm{Am}] + [\mathrm{AmX}] + [\mathrm{AmX_2}] + \ldots \tag{22.6}$$

where $[\mathrm{Am}]_\mathrm{T}$ is the total analytical concentration of americium. Dividing by [Am] gives:

$$[\mathrm{Am}]_\mathrm{T}/[\mathrm{Am}] = [\mathrm{Am}]/[\mathrm{Am}] + [\mathrm{AmX}]/[\mathrm{Am}] + [\mathrm{AmX_2}]/[\mathrm{Am}] + \ldots \tag{22.7}$$

$$[\mathrm{Am}]_\mathrm{T}/[\mathrm{Am}] = 1 + \beta_1[\mathrm{X}] + \beta_2[\mathrm{X}]^2 + \ldots \tag{22.8}$$

and

$$\alpha_n = \beta_n[\mathrm{X}]_n([\mathrm{Am}]/[\mathrm{Am}]_\mathrm{T}) \tag{22.9}$$

where α_n is the fraction of all americium in the form of AmX_n. With these equations, β_n's, and a given value of $[\mathrm{Am}]_\mathrm{T}$, we can calculate the concentration of each species for any value of [X]. In Figure 22.9 the speciation of Am(III) as hydroxide and carbonate complexes is shown as a function of pH. The concentration of CO_3^{2-} is based on the

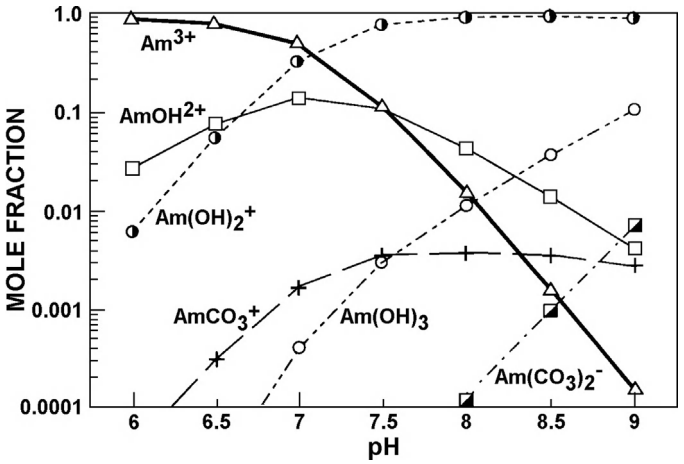


Figure 22.9 Results of speciation calculations for Am(III) in natural water.

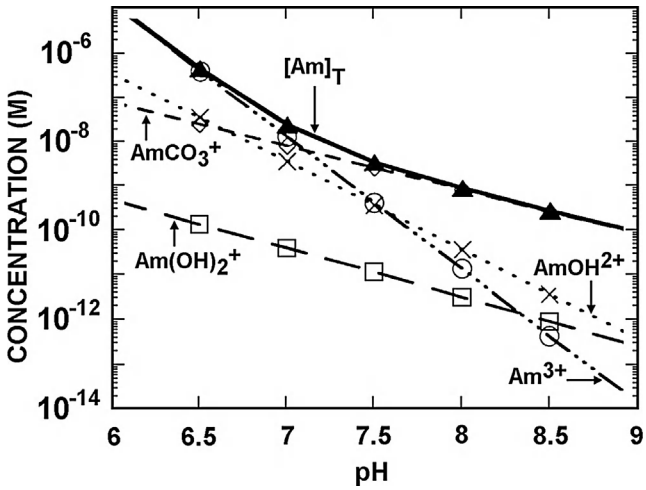


Figure 22.10 Results of speciation calculations for Am(III) in natural water, taking into account the low solubility of $Am(OH)_3$.

atmospheric partial pressure (3.2×10^{-4} atm) and the resultant concentration of CO_3^{2-} at different pH values in a natural water.

If the solubility is limited by a solid phase for which the K_{s0} is known, the actual value to be expected for each species can be calculated by assuming no over-saturation. These can be included in the mass balance equation to predict the total minimum solubility. If $\log K_{s0} = -28.9$ is assumed for $Am(OH)_3$, Figure 22.10 shows the results of such a calculation for the solubility of americium in the same system as in Figure 22.9, but now including the effect of the limited solubility of $Am(OH)_3$.

We have stated that Pu(V), as PuO_2^+, is the dominant dissolved species while $Pu(OH)_4$ is the solubility limiting precipitate. The redox reaction can be written as:

$$PuO_2^+ + 4H^+ + e^- = Pu^{4+} + 2H_2O \tag{22.10}$$

The E^0 value for the IV/V pair is 1.17 V. Using the equilibrium expression with the Nernst equation, assuming the redox potential in a fresh water lake to be 0.4 V, we obtain at pH 7:

$$\left[PuO_2^+\right] / \left[Pu^{4+}\right] \approx 10^{15} \tag{22.11}$$

From the value of $\log K_{s0} = -56$ for $Pu(OH)_4$(am) we can calculate that $[Pu^{4+}]$ in a solution of pH = 7 in contact with solid $Pu(OH)_4$ is 10^{-28} M. This gives us a value of $10^{15} \times 10^{-28} \approx 10^{-13}$ M for the expected concentration of PuO_2^+ in this solution.

A number of geochemical modeling codes have been developed which use such speciation and solubility equilibrium equations to calculate the concentration of different species of a metal ion as well as its net solubility in various waters (common codes are PHREEQC and EQ 3/6). The results from such calculations are as good as the equilibrium constants or the thermodynamic values used in the calculations. However, most important, the calculations must include the equilibrium equations for all species present which may contribute significantly to the solution phase concentrations and all solids which can provide the limiting solubility to the solution species. Furthermore, the degree of oversaturation with regard to each solid phase must be prescribed in order to calculate a realistic speciation. These modeling codes are presently based on the assumption that the natural systems are all at equilibrium whereas in nature this may not be true. Many systems are kinetically controlled and are often in a steady state, but not in true equilibrium. In these cases, perhaps the majority of the systems, the equilibrium modelling codes cannot accurately describe the actual conditions, but may provide a set of limiting species, approximate relative concentrations and baseline net solubilities. A further complication arises in assessing the role of colloids, and of sorption which may reduce the concentration of soluble species below that estimated for the least soluble solid phase. On the other hand, sorption on suspended colloids may also increase the total concentration (dissolved plus amount in colloids, e.g. Pu in sea water §22.5). In general, the equilibrium code calculations can easily give lower limit values of maximum solubilities by assuming no oversaturation. Such calculations are valuable in waste management risk assessment since, if the lower limit solubilities from the equilibrium calculations fall well below the accepted safety limits even when assuming reasonable degrees of oversaturation, it is very likely that the actual total concentrations will also be below the acceptable limits, cf. Fig. 22.11, Table 22.10 and §22.10.

22.7.1. Calculated species in solution

The diversity of reactions which actinides can undergo in natural waters is presented schematically in Figure 22.11. Complexation by anions such as hydroxide, carbonate, phosphate, humates, etc. determine the species in solution. Sorption to colloids and suspended material increases the actinide concentration in the water while precipitation of hydroxides, phosphates, carbonates, and/or sorption to mineral and biological material limit the amount in the solution phase.

In natural oxic waters, americium is present in the trivalent and thorium in the tetravalent state while uranium is hexavalent, UO_2^{2+}. The total concentrations of uranium and thorium in surface sea water are $1.1-1.5 \times 10^{-8}$ and $\sim 2.5 \times 10^{-12}$ M, respectively in both the Atlantic and Pacific oceans. The amount of uranium associated with particulate matter in the water is small. By contrast, for tetravalent ^{232}Th (originating from thorium minerals), about 50% is bound in aluminosilicate particles

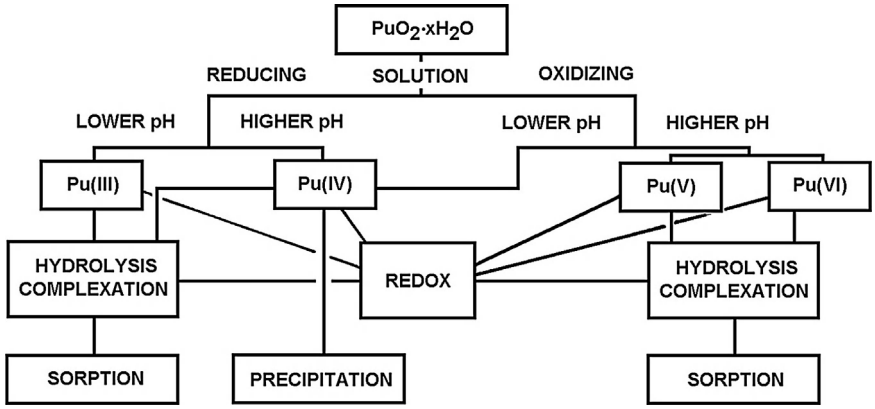


Figure 22.11 Speciation diagram for the range of reactions to be considered in studying the environmental behavior of plutonium.

and 50% is dissolved (e.g. passes through a 1 μm filter). For ^{228}Th and ^{230}Th (radiogenic decay products), about 90% is found in solution. Sorption of ^{234}Th tracer is a reversible process, possibly due to an organic material coating on the surface with increased sorption as the particles age. In such studies, solubility is defined as passage through a 0.45 μm filter. Speciation calculations show typically that thorium occurs as a hydrolyzed species whereas uranium is present in surface waters, at low in Ca^{2+}, normally as the $UO_2(CO_3)_2^{2-}$ and $UO_2(CO_3)_3^{4-}$ species. In neutral waters of very low carbonate content, speciation calculations indicate that uranyl cations hydrolyze to form oligomers for total uranyl concentrations as low as 10^{-7} M.

Neptunium, in oxic waters, is present in the pentavalent state. The hydrated cation is calculated to be the dominant species for pH$\leq$8 unless the free carbonate concentration exceeds approximately 10^{-4} M in which case $NpO_2(CO_3)_n^{1-2n}$ is more common.

The rather similar values for the plutonium reduction half reactions at pH 8 indicate that plutonium may exist in oxic waters in more than one oxidation state. The reduction potential at pH 8 of the Pu(III)/Pu(IV) couple indicates that Pu(III) is unlikely to exist in oxic waters in the absence of a reductant, but may be present in anoxic waters. Each oxidation state of plutonium differs in chemical behavior from that of the other states so modelling the geochemical behavior of plutonium must include the correct oxidation state, or states, of plutonium which are likely to be present in a particular system.

In principle, the calculation of the oxidation states of plutonium requires knowledge of the redox potential, *E*h, of the aqueous phase. However, the *E*h measured with a certain type of electrode may not be the potential for the particular redox couple with which the plutonium reacts. One of the reasons for this is that the

Table 22.8 Plutonium oxidation states (in % of total Pu) in natural waters

State	Pu(III+IV)	Pu(V)	Pu(VI)
Rain	34	66	
Mediterranean	42	58	
Irish Sea	23	77	0
Pacific (I)	39	52	9
Pacific (II)	40	46	14
Lake Michigan	13	87	0

Table 22.9 Equilibrium Constants for Am(III)

Species	$\log\beta_n$	Species	$\log\beta_n$
$Am(OH)^{2+}$	6.44	$Am(CO_3)^{+}$	5.08
$Am(OH)_2^{+}$	13.80	$Am(CO_3)_2^{-}$	9.27
$Am(OH)_3$	17.86	$Am(CO_3)_3^{3-}$	12.12

Eh-electrode usually catalyses the reaction rate for its specific redox couple. For example, in surface sea water, the measured *E*h is about 0.8 V and is due to the O_2/H_2O couple. In the log *E*h versus pH diagram, Figure 22.6, the area of existence of plutonium in different oxidation states, including the effects of hydrolysis and carbonate complexation, is marked by roman numerals. From that diagram, it can be seen that Pu(VI) would be the predominant state in solution in the ocean. In fact, the predominant species is Pu(V). Table 22.8 summarizes the reported oxidation state distribution in several natural systems.

Like NpO_2^+, PuO_2^+ has a low tendency to hydrolysis and complexation and is much less likely to be sorbed to solid surfaces and on colloidal particles than the Pu species in other oxidation states. As a consequence, plutonium can be expected to migrate most rapidly if it is in the pentavalent oxidation state. The total solubility is limited by the formation of the highly insoluble $Pu(OH)_4$. The sorption of hydrolyzed Pu(IV) in neutral water on mineral and organic-coated surfaces is accountable for the very low concentrations of dissolved Pu even in the absence of $Pu(OH)_4$(am) or PuO_2(c). Desorption is accomplished only by strong complexing and/or redox reagents. For example, citrate extracts little plutonium from soil, but a combination of citrate and the redox agent dithionite provides good extraction. The intractable nature of $Pu(OH)_4$ and its strong tendency to sorb on surfaces is a dominant and often controlling feature of plutonium geochemistry.

Silicates and humic substances present in natural waters form colloids and pseudocolloids with which actinides can react. The pseudocolloids formed by humic substances in ground waters have been shown to be efficient scavengers of americium and plutonium.

22.8. NATURAL ANALOGUES

The geochemical modeling calculations based on measured equilibrium data are of primary importance in the safety assessments of proposed nuclear waste repositories (and should be important for other waste repositories as well). Another useful tool in such assessments is the data from studies of natural analogue sites.

The modeling calculations use data from laboratory and field studies which have been obtained over the last few decades. These data are used in the codes to predict the solubilities, and nuclide migration of material which might be released from nuclear repositories over thousands and, even, hundreds of thousands of years. It is not possible to demonstrate rigorously that the models used are accurate as they may simplify the natural system, use incorrect data or misrepresent (possibly, ignore) important processes which occur over very long time periods. However, some validation of the models and data used in the modeling calculations can be obtained from careful comparison of calculated values with those measured at appropriate geologic sites, known as a natural analogues.

The natural analogue sites are areas in which uranium ores have been present for geologic time periods. In most cases, these sites have not been affected by human activities, so the record of geologic, long term effects are well preserved. A number of such sites are being studied around the world; studies from a few sites of different characteristics are reviewed here.

From studies of granitic sites, the dominant role of fractures and fissures in the transport of fluids has been convincingly demonstrated. Thus any model of water in a granitic site must include both advection which is dominant in the fractures and fissures, and diffusion which is important in regions of highly altered rocks ("alteration rims"). In clays, mass transport seems to proceed primarily by ionic and molecular diffusion although some fluid mass transport occurs at discontinuities in the formation.

Often, the mobilization of many elements (e.g. U, but not Th, in crystalline rocks) is correlated with the flow of oxidizing water. The mobilization and fixation of uranium involves complexation, redox and retention on minerals via adsorption, and ion exchange. In clay media, the redox potential is strongly buffered if significant amounts of organic substances are present.

In a section of the Pocos de Caldas (Brazil) formation, most of the thorium and the rare earths, and, to a lesser extent, the uranium, is associated with goethite (FeOOH) particles and transport by organic colloids is much less important. This region is reducing, as shown by the presence of Fe(II). However, in another region of this formation, the thorium and rare earths have a much higher mobility and are associated with organic (humic) colloids. At an analogue site in Scotland, the flow from the ore has passed into a peat bog in which the uranium is associated predominantly with humic material while the Th is found on Fe/Al oxyhydroxide colloids and particles. In many